EXPLORING THE ASSOCIATION OF VAGINAL MICROBIOME AND PREECLAMPSIA USING A NETWORK-BASED APPROACH

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Introduction

- Women with vaginal microbiome dysbiosis and a high prevalence of pathogenic microbes are more likely to develop preeclampsia, a fatal pregnancy-related complication for both the mother and baby.
- We built a microbial interaction network of the microbes that inhabit the vaginal microbiome to understand the factors that contribute to preeclampsia using an in-silico approach.

Methodology

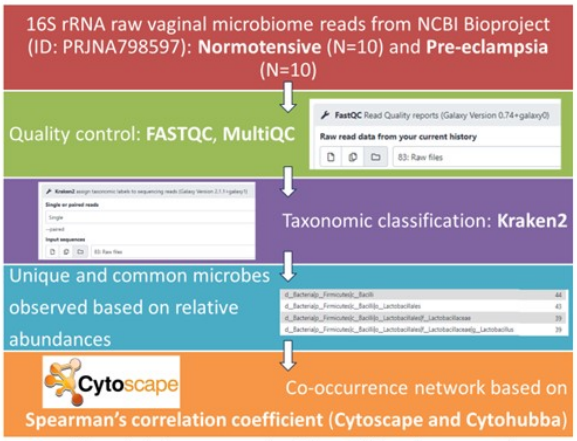

Fig. 1. Taxonomic Classification and Correlational analysis workflow using Galaxy

Results

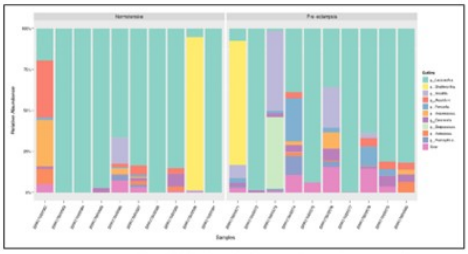

Fig. 2. Microbial relative abundances in Normotensive and Pre-eclampsia vaginal microbiome samples

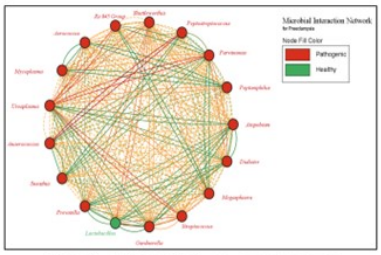

Fig. 3. Microbial co-occurrence network. Green lines show strong positive interactions ($0.5<\rho<1$)

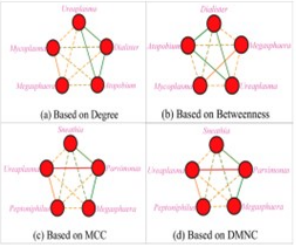

Fig. 4. Hub microbes identified using Cytohubba (Based on Degree, Betweenness, MCC, DMNC)

Conclusion

- **<u>Unique to Pre-eclampsia samples</u>**: *Megasphaera, Aerococcus, Streptococcus,* and *Peptostreptococcus* (**All pathogenic**).
- **<u>High in abundance in Pre-eclampsia samples</u>**: *Aerococcus, Atopobium, Gardnerella, and more* (**All pathogenic**)
- **<u>Lower abundance in Pre-eclampsia samples</u>**: *Lactobacillus*
- **Strong** positive interactions between pathogenic microbes, weak interactions with *Lactobacillus*.
- **<u>Common Hub microbes</u>:** *Ureaplasma*, and *Megasphaera (***Both pathogenic)**

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